The Last Clue

Checkpoint 2: Transformation and Animation (pure C++)

Each page shows four key frames (t = 0, 1/3, 2/3, 1). The full GIFs are in docs/gifs/ in the repository.

Group members - CP2 work

Caleb Kipchirchir - shear and reflection, our own animated GIF encoder (palette + LZW), integration

Latiffa - animation core (lerp, easing, keyframes), evidence board animation, scene transitions

Alvin Kimani - detective walk cycle (hierarchical transforms, sprite scaling), door/drawer opening

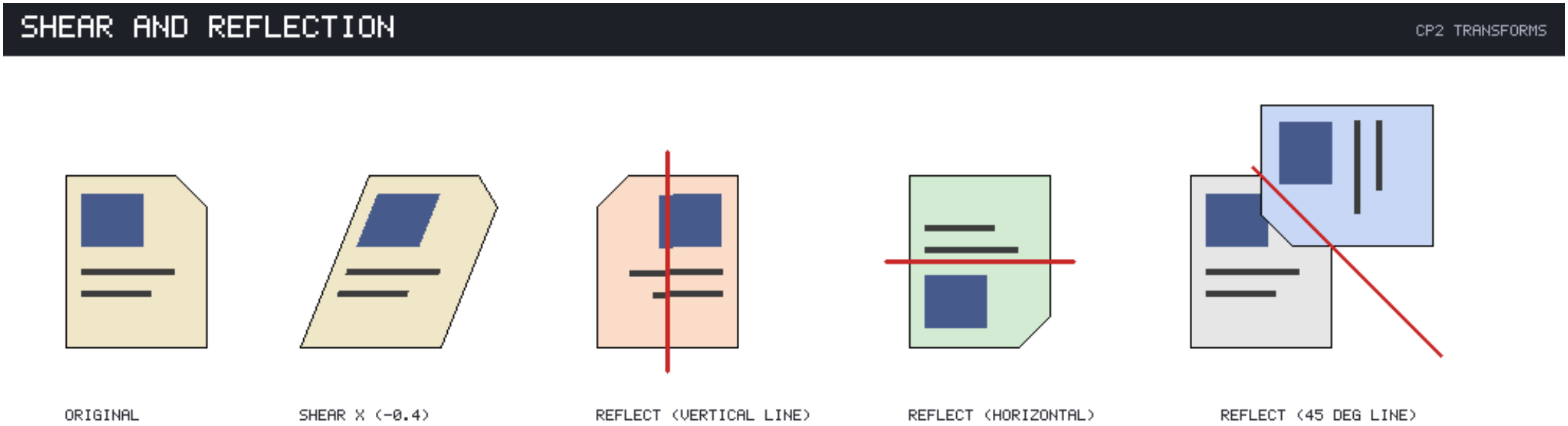
Sudheysi Ali - safe combination + door, torn letter reassembling, magnifier zoom

Repository

https://github.com/C-Sawe/the-last-clue

1. Shear and reflection

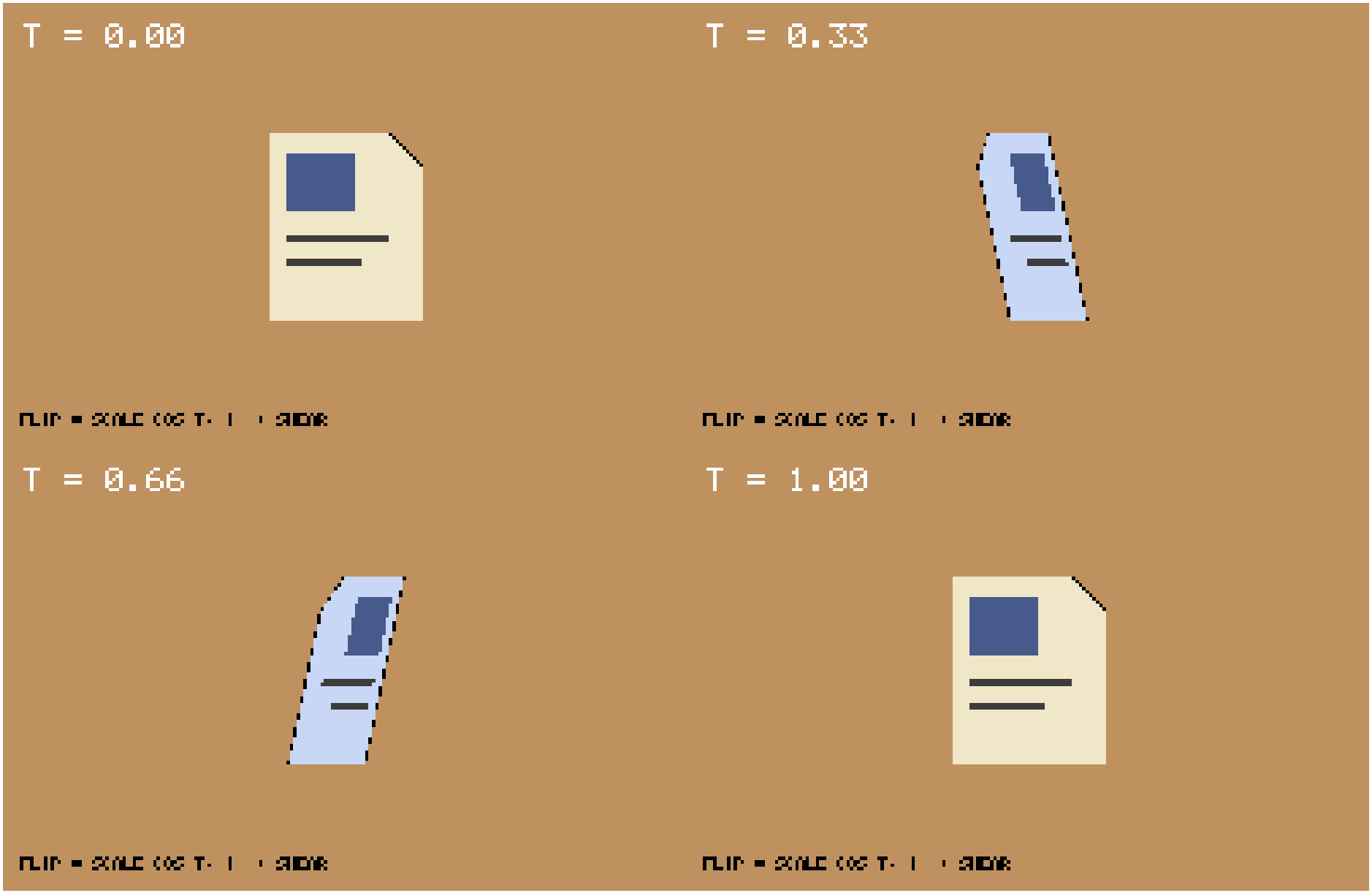
Implemented by Caleb



Shear x' = x + sh*y; reflection in any line through p at angle a = T(p) R(a) S(1,-1) R(-a) T(-p), one 3x3 matrix.

2. Animated card flip (our GIF encoder)

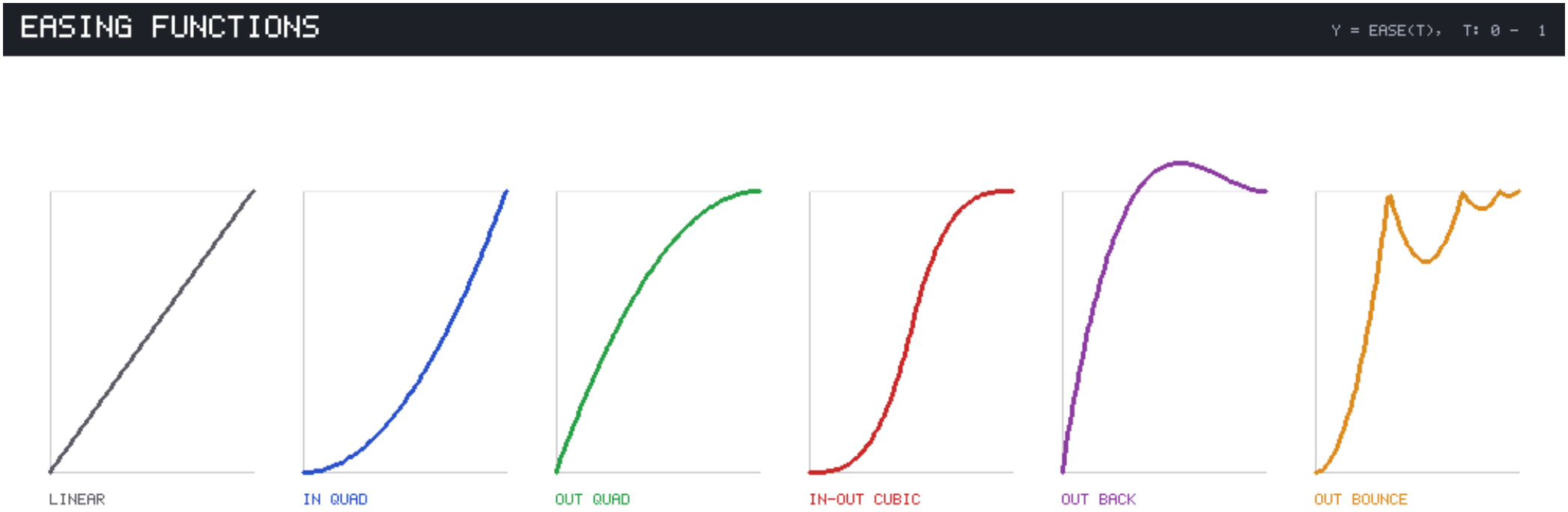
Implemented by Caleb



Scale(cos t, 1) with a small shear. Every animation in CP2 is saved by our own GIF writer: shared 256-colour palette + LZW compression.

3. Easing functions

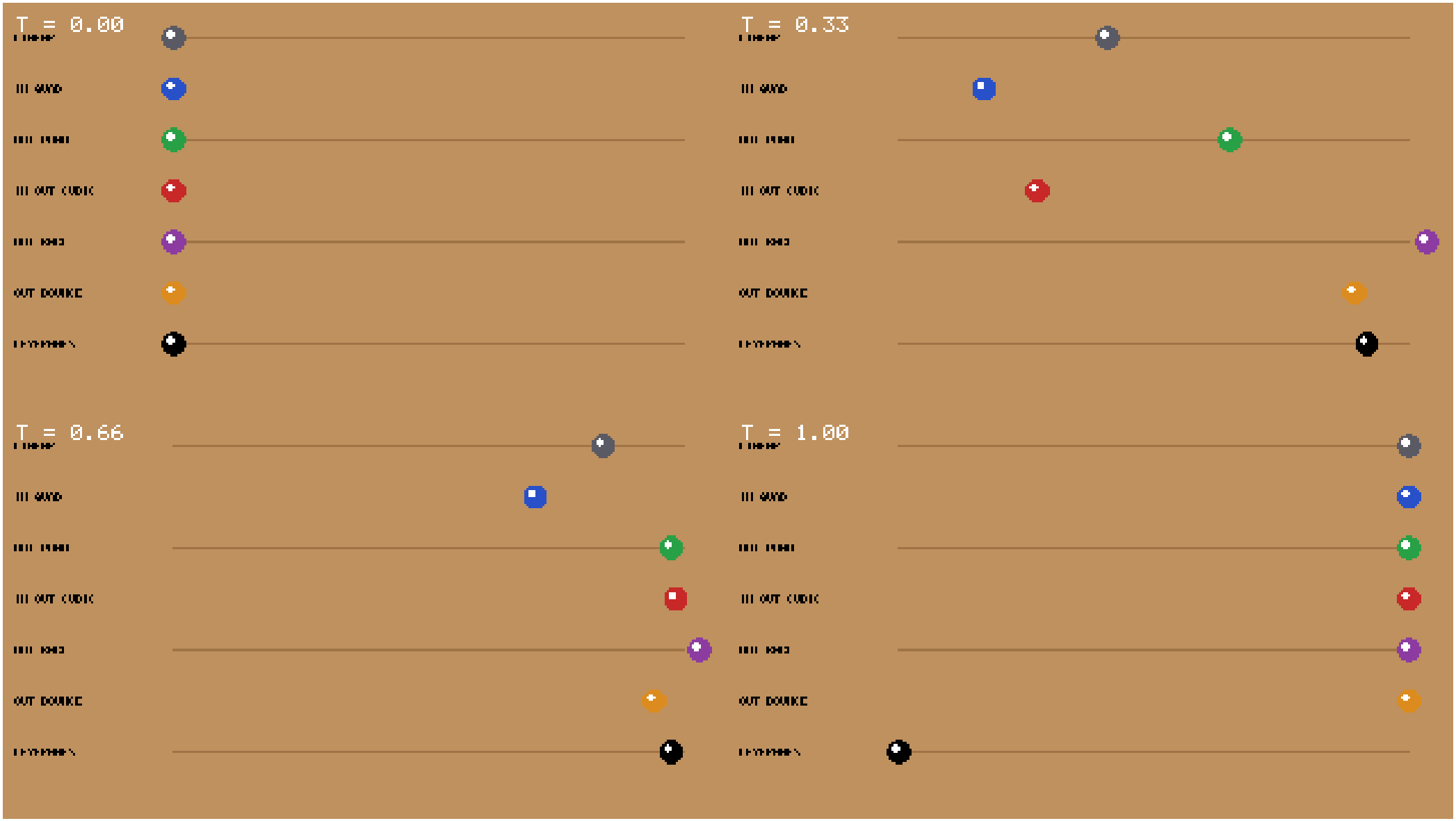
Implemented by Latiffa



y = ease(t) for Linear, InQuad, OutQuad, InOutCubic, OutBack (overshoot) and OutBounce.

4. Easing + keyframe tracks

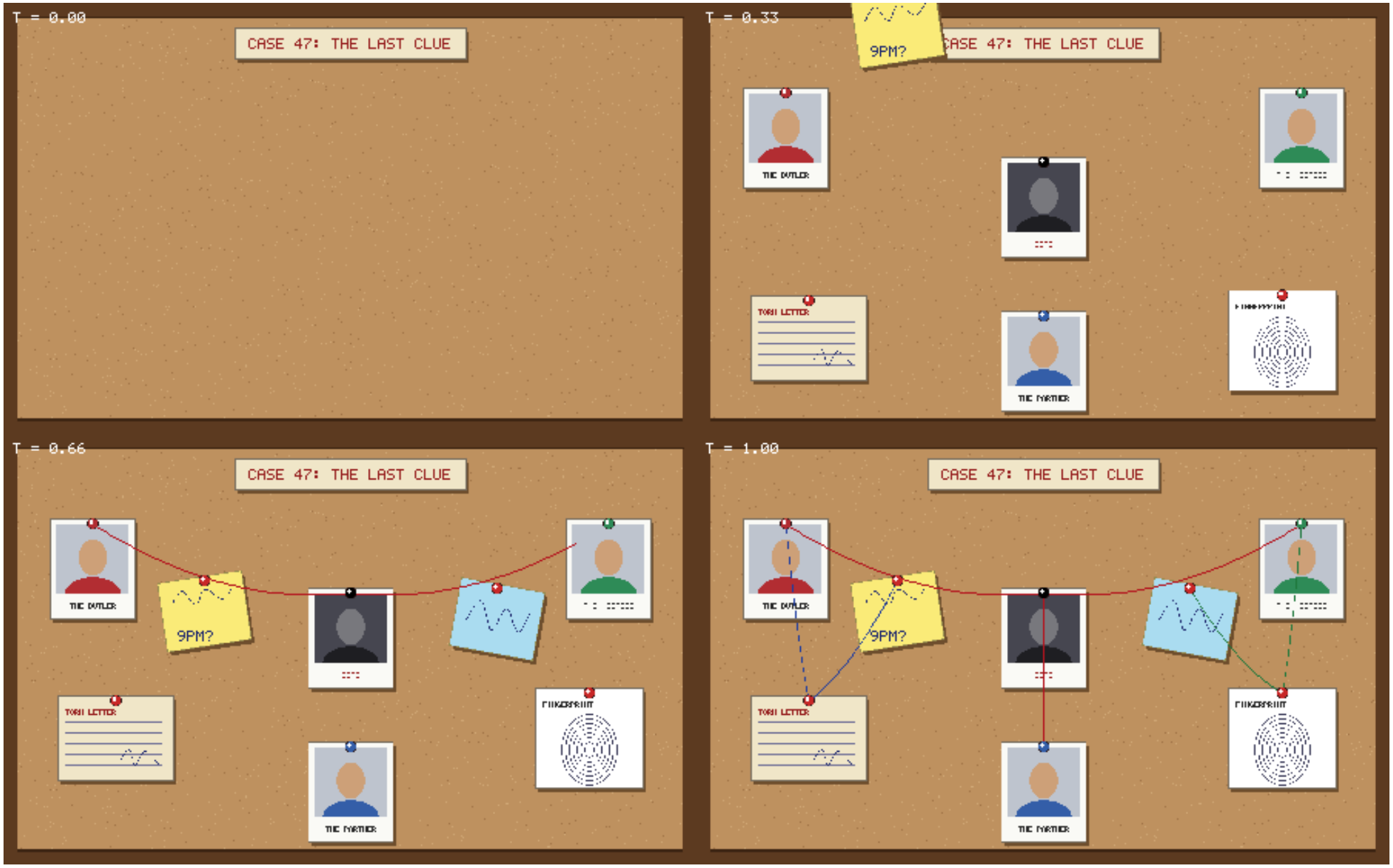
Implemented by Latiffa



Seven pins travel the same distance with different easing; the black one follows a keyframe Track (bounce in, hold, return).

5. Evidence board animation

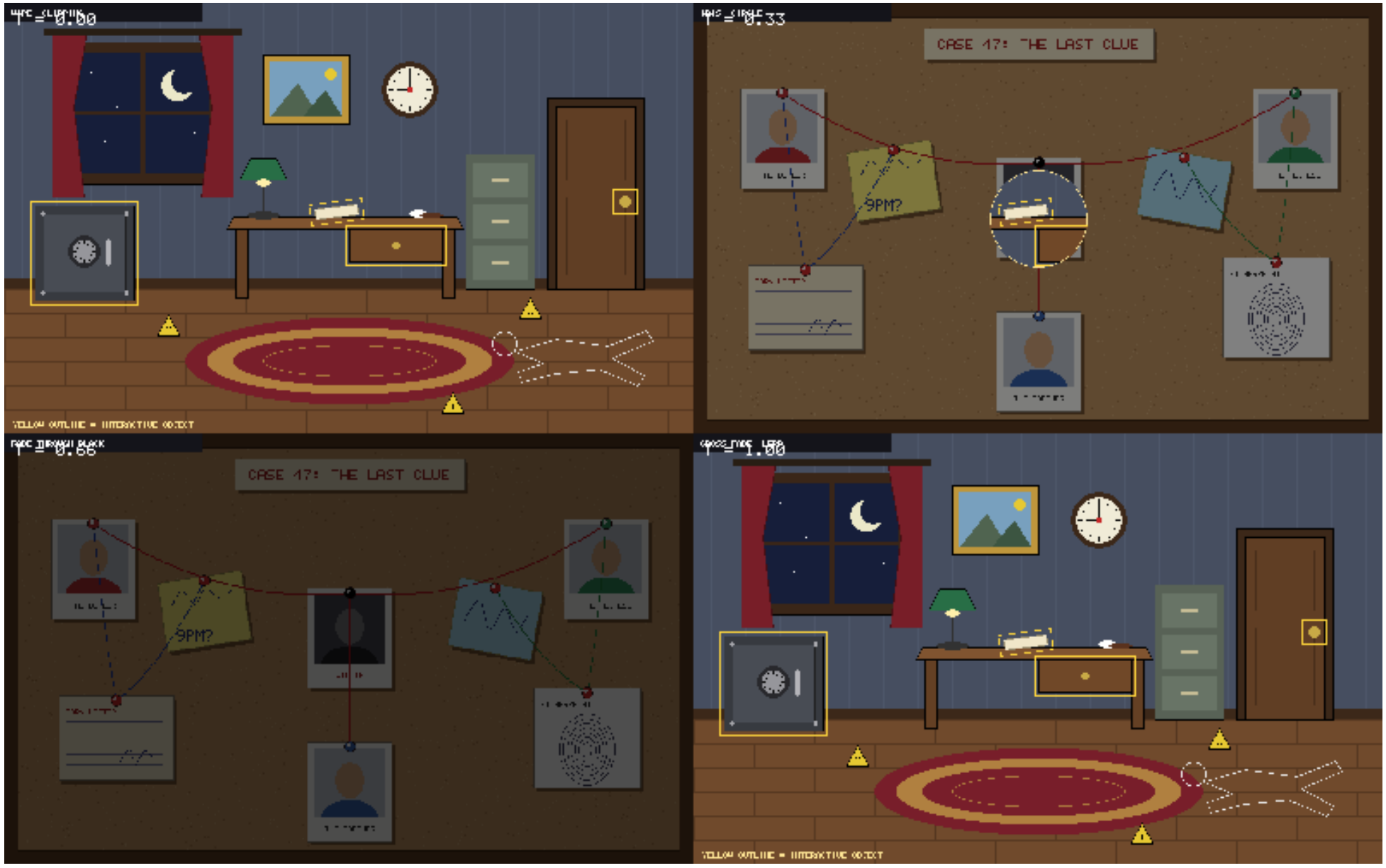
Implemented by Latiffa



Cards drop in one after another (translation, OutBack easing, staggered phases); strings are then pulled pin to pin by drawing a partial Bezier curve, t: 0 -> tEnd.

6. Scene transitions

Implemented by Latiffa



Wipe (growing clip window, Sutherland-Hodgman + Cohen-Sutherland edge), iris (circle test), fade through black and cross-fade (per-pixel colour lerp).

7. Detective walk cycle

Implemented by Alvin



Hierarchical transforms: legs rotate about the hips and arms about the shoulders (sin of the step phase); the sprite is scaled into the room with one matrix and drawn by inverse mapping.

8. Door and drawers opening

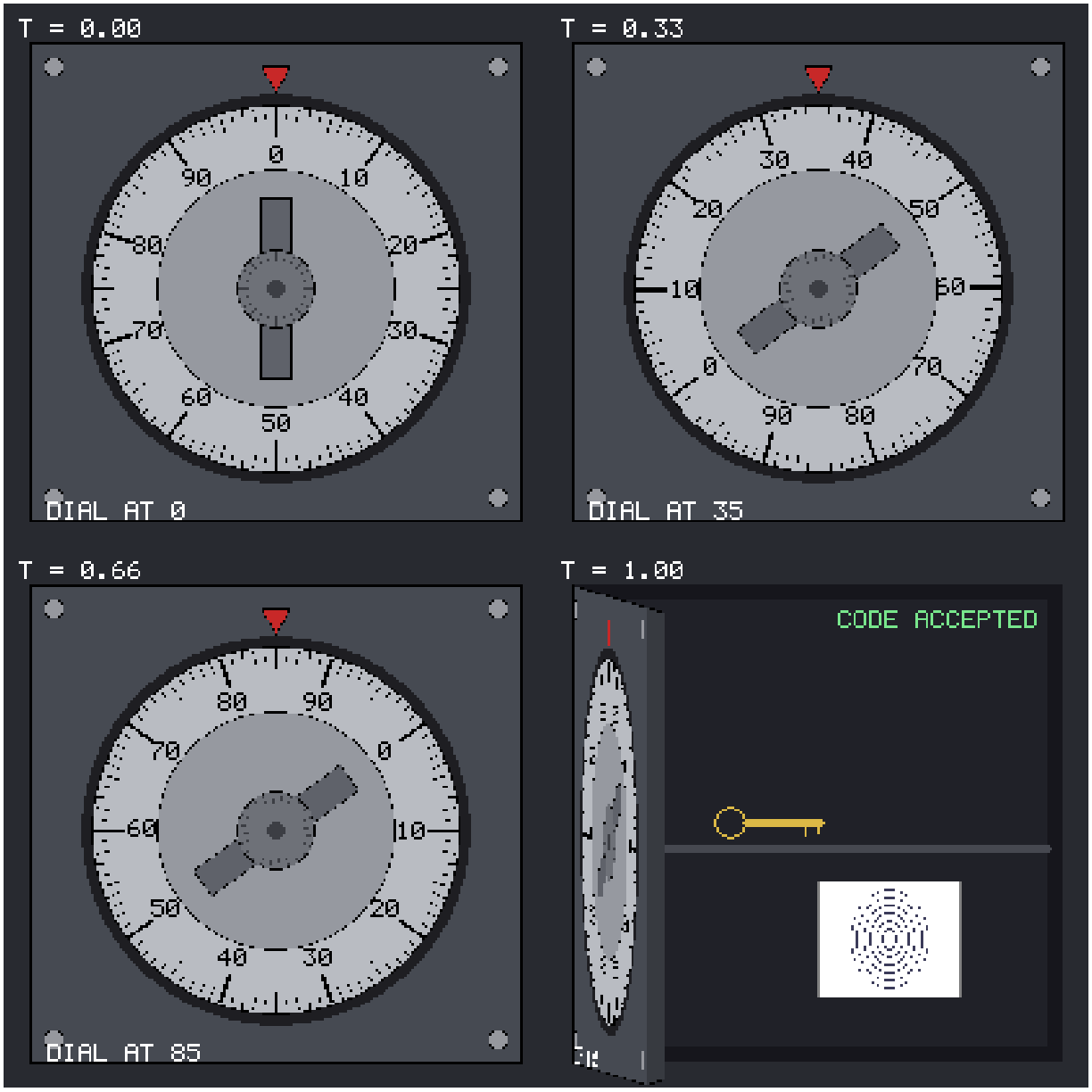
Implemented by Alvin



Door leaf: scaleAbout + shearAbout its hinge (fake perspective) revealing the hallway; drawers: translate + slight scale (closer = bigger). Keyframe tracks with easing.

9. Safe combination and door

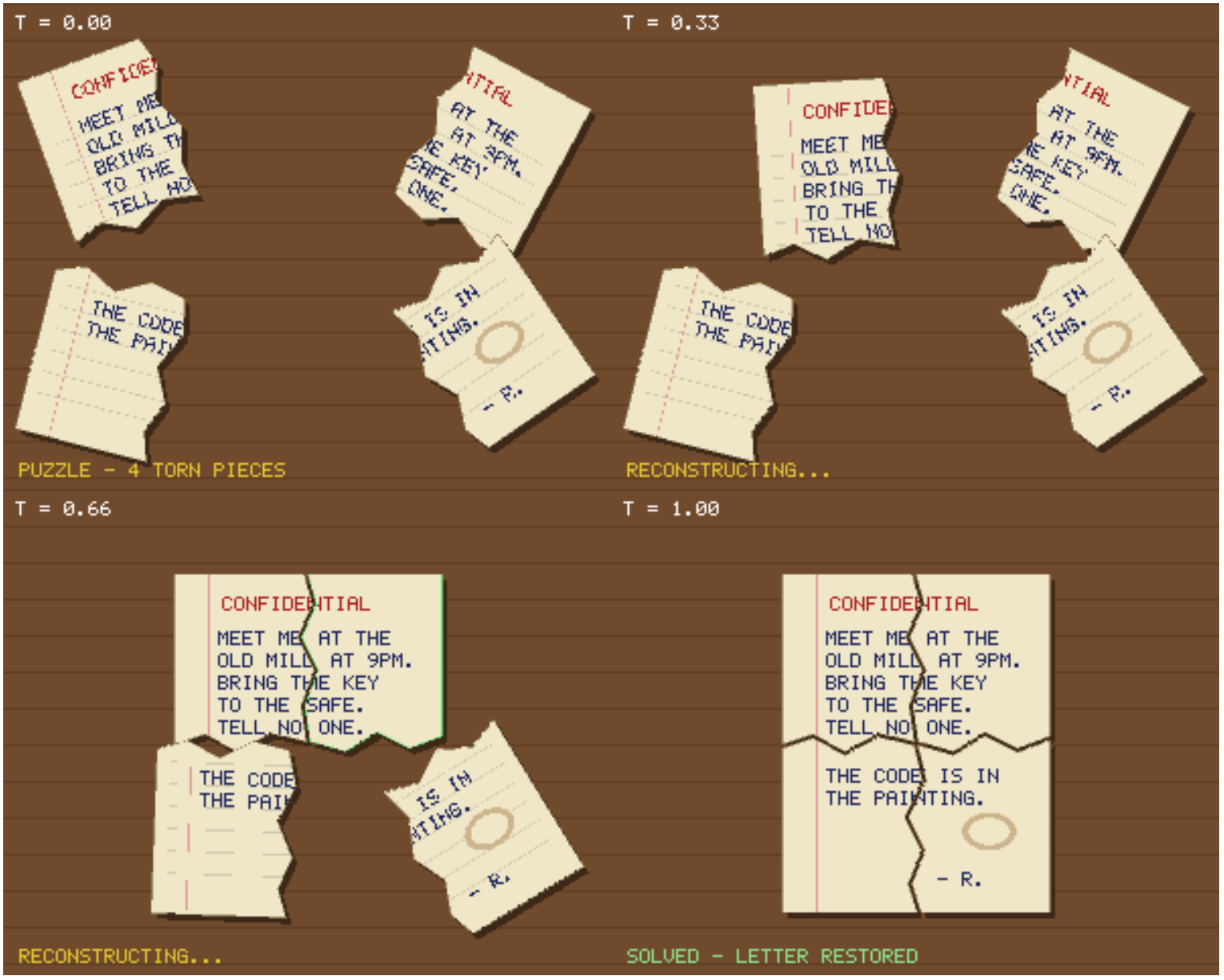
Implemented by Sudheysi



The dial follows keyframes 37 - 12 - 85 (rotation about the dial centre, eased, with pauses); the door plate then swings about its hinge (scale + shear), revealing the key and fingerprint card.

10. Torn letter reassembles

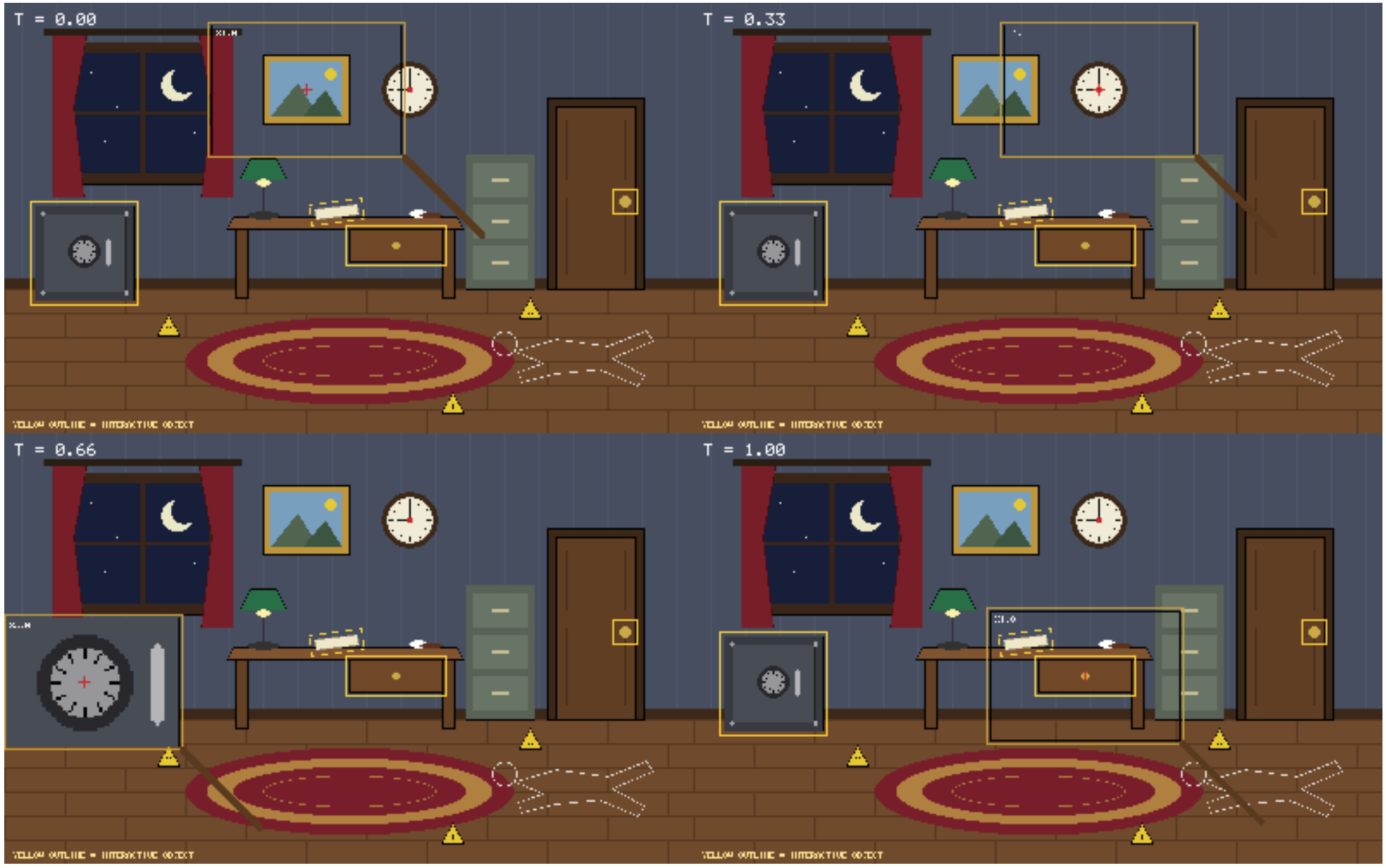
Implemented by Sudheysi



Each piece interpolates from its scattered pose to the solved pose: M(u) = T(lerp(d,g,u)) R(lerp(a,0,u)) T(origin), staggered and eased; a green outline flashes when it snaps.

11. Magnifier zoom

Implemented by Sudheysi



The lens glides between clues; inside it the scene is scaled about the focus point (inverse mapping), the lens is clipped to the screen (Sutherland-Hodgman) and the handle line too (Cohen-Sutherland).